

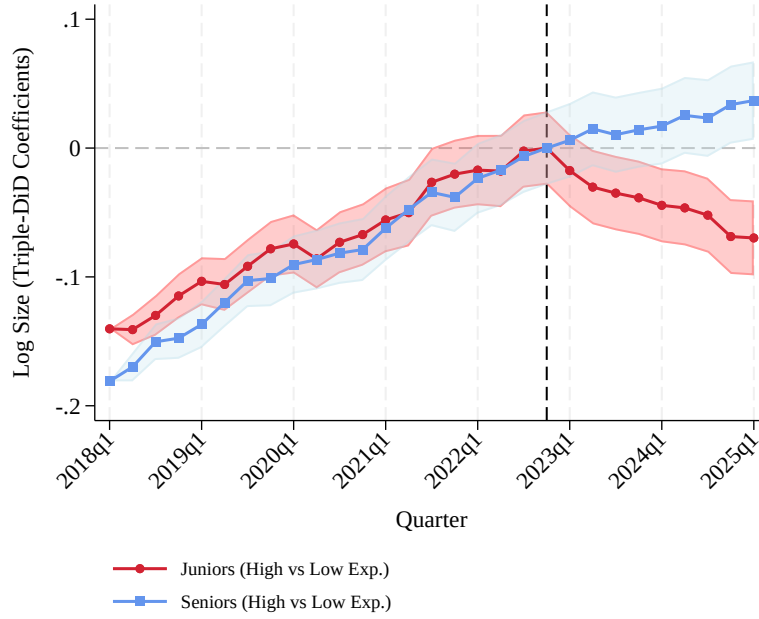


Figure 6: Triple-Difference by Occupational Exposure—Juniors vs. Seniors

Notes: This figure reports the estimated coefficients β_j^s from Equation 12, estimated separately for juniors (red) and seniors (blue). Each series captures the evolution of log employment in high-exposure occupations relative to low-exposure occupations at adopters relative to non-adopters, conditional on firm-by-time and industry-by-exposure-by-time fixed effects. Details on the exposure measure are provided in Section 4.1. Shaded bands are 95% confidence intervals. Standard errors are clustered at the firm level.

estimates the DiD specification within tiers of juniors' educational background, measured by the prestige of their alma mater. The relative decline in junior employment is U-shaped across tiers: largest for juniors from mid-tier institutions and smallest for juniors from the most and least selective institutions.

5.2.4 Discussion: The Timing of the Junior Employment Decline

The early and pronounced decline in the triple-difference coefficients—starting to emerge in 2023Q1, shortly after the release of ChatGPT—may appear surprising, given that automation effects typically unfold more gradually. One plausible explanation is that the rapid diffusion of GenAI tools at the end of 2022 constituted a discrete shock to firms' expectations about future automation. In response, firms may have adjusted in a forward-looking manner by scaling back hiring for junior roles they anticipated would be automated in the near future. Such preemptive adjustments could be optimal for firms if

A.3 Heterogeneity by Educational Background

This appendix examines whether the decline in junior employment varies by workers' educational background, as referenced in Section 5.2.3. We first describe the construction of the school-quality variable, then present the heterogeneity results.

A.3.1 Construction of the School-Quality Variable

To capture educational background, we construct a position-level measure of school quality. For each position, we assign the institution where the worker most recently studied, provided that the education ended no later than one year after the job start. If no such record exists, we use the first recorded institution, provided it began before the job start date. Each institution is then assigned a GPT-4-based quality rating on a five-point scale, produced using OpenAI's GPT-4o-mini model (see Supplemental Appendix A.11.2 for the full prompt). A score of 1 corresponds to Ivy League and other globally elite institutions; 2 to highly respected international universities; 3 to strong national or regional schools; 4 to lower-tier but standard institutions; and 5 to minimally selective or non-accredited institutions.

A.3.2 Results

We re-estimate Equation 34 for junior employment separately within each of the five school-quality tiers. For clarity of presentation, we use a single post-treatment indicator for quarters after 2023Q1 instead of a full set of time dummies.

The results, shown in Figure A.12, reveal a U-shaped pattern. Juniors from tier-3 universities experienced the steepest relative declines in employment, while those from tiers 2 and 4 also saw significant reductions. Juniors from tier-1 and tier-5 universities experienced the smallest decline.³³ Figure A.13 reports the time-series version of these estimates, with a full set of time dummies in place of the single post-2023 indicator.

To help interpret this pattern, Figures A.14 and A.15 show average predicted salaries³⁴

³³In the analysis, we exclude positions where the worker's most recent educational entry is reported as a high school (approximately 5 percent of positions). Including these positions, based on high school quality, has virtually no effect on the results.

³⁴Actual position-level salaries are unavailable. However, Revelio Labs provides a predicted salary

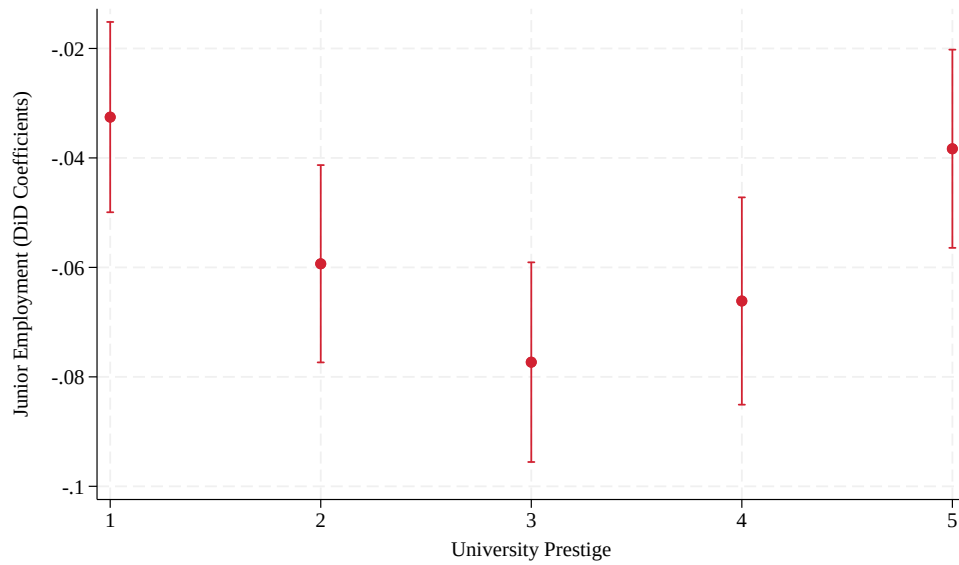


Figure A.12: Heterogeneity by School Quality

Notes: Each bar reports the estimated coefficient from re-estimating Equation 34 for junior employment separately within each of the five school-quality tiers. The specification replaces time dummies with a single post-treatment indicator for quarters after 2023Q1. Estimation begins in 2018 to exclude early-sample trends. Standard errors are clustered at the firm level.

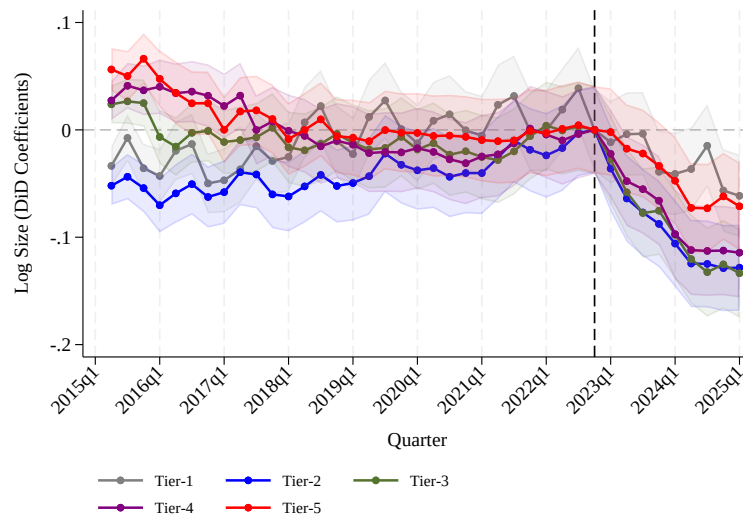


Figure A.13: DiD by Educational Background—Time Series (Juniors)

Notes: This figure plots β_j from Equation 34, estimated separately for each of the five school-quality tiers. Standard errors are clustered at the firm level.

variable imputed using position-specific information. For more details, see: <https://www.data-dictionary.reveliolabs.com/methodology.html#salary>.

A.28 The Junior Hiring Decline—Heterogeneity by Sector

This appendix examines heterogeneity in the decline in junior hiring by sector. For this, we re-estimate Equation 15 for junior hires separately by sector. Results are presented in Figure A.30. Across all sectors, adopting firms exhibit a sharp and statistically significant relative decline in junior hiring after 2023Q1, while senior hiring remains stable or increases slightly. This pattern indicates that the contraction in junior hiring is broad-based across industries and not driven by any single sector disproportionately reducing demand for junior workers.

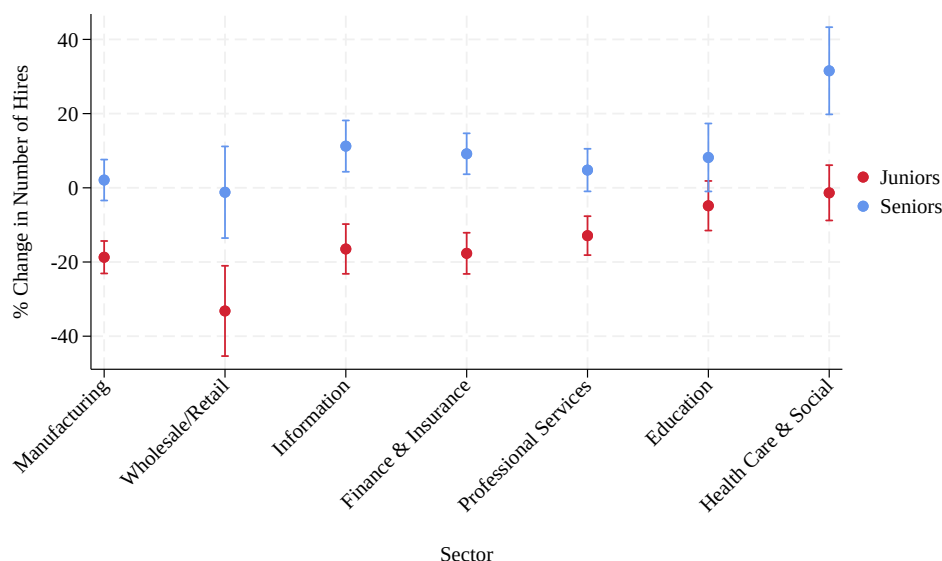


Figure A.30: Estimated Effects of Generative AI Adoption on Hiring by Sector

Notes: Sectors correspond to the following NAICS classifications: Manufacturing (31–33), Wholesale/Retail (42, 44–45), Information (51), Finance & Insurance (52), Professional Services (54), Education (61), and Health Care & Social Assistance (62). All coefficients are normalized by the pre-2022 average number of hires in each sector-adoption group. Standard errors are clustered by firm.